尝试调整 没有思路

Personal Statement

第一段可以考虑补充一下，1）下面三个sections的简介，2）你学习成绩（archivements）的简介。目前本段最后一句太宽泛。

such as 后面要加多个例子，上次讲过

改为especially

I am applying for the PhD program in Physics at National University of Singapore to pursue my interest in exploiting first-principles modeling methods such as density functional theory (DFT), combined with data-driven techniques like machine learning (ML) to help understand, discover, and design material systems. I believe my research experience and training will enable me to excel in the PhD program.

逻辑上有点突兀，前面应该加一两句展示你的成绩的。

独立一行做小标题，下同。上次讲过，是有什么顾虑吗？ 最后两段不好加标题

好像挖掘不出什么 只是联系了一下 没什么思路

Research Experience in first-principles studies My journey into first-principles studies began during my sophomore year, when the COVID-19 pandemic kept us at home and prevented access to physical laboratories. ~~Fortunately~~, Prof. Zhang Peng offered me the opportunity to engage in DFT studies on the vibrational properties of ice phases. The DFT approach enabled effective computational experiments with sufficient accuracy. Building on this advantage, I transformed an old desktop computer into a computing workstation and started working on the Phonon Density of States (PDOS) for ice III. Over two months, I explored various modeling methods and computational parameters, ultimately achieving agreement between calculated PDOS and experimental data. As the far-infrared spectrum holds great potential due to emerging terahertz technology, my next goal is to explain the origin of this spectral range. I partnered with a graduate student to integrate DFT-simulated vibration normal modes with lattice dynamic analysis. Through a comparative analysis with ice IX, we proposed that the overlapping of two intrinsic hydrogen bond vibration modes, namely two-bond modes and four-bond modes, has led to the flat translational band in ice III. Furthermore, we attributed the splitting of degenerate vibrational modes to the hydrogen-disordered structure. This work led to my first published paper.

1）covid和Prof.Zhang的机会写成一句话，转折；2）Prof.Zhang为何会给你这个机会呢？

这是个重要信息吗？想说你遇到了什么困难又展示了你的什么能力吗？未能清晰表达

这到底是发生在过去还是未来？过去 已改

为什么会和ta合作呢？

已改 At the graduation ceremony of Shandong University

这句话引用得不错，但表达不够引人入胜，尝试调整。比如，"在毕业典礼上，我被….“

Upon graduating from Shandong University, our dean told us that "academic research is not exclusive to a few brilliant minds; anyone who dedicates 10 years to a field can achieve success" Encouraged by this advice, I decided to put it into practice. During my master program at NUS, I continued to work on first-principles studies under the supervision of *Prof. Quek Su Ying*. My master project aims to find promising quantum point defect in 2D WS_2. Specifically, the project focused on 4d and 5d transition metal (TM) substitutional impurities in multiple charge states. I developed a VASP screening workflow to identified deep-level defects with spin-triplet ground state. Performing large-scale calculations on the NUS HPC cluster has allowed me to use and hone skills in automation (Bash and Python), high-throughput processing (VASPKIT), job scheduling (PBS), and parallelization. As a result, three candidates were chosen for future evaluation on their magneto-optical properties. Additionally, I presented a statistical overview on the point groups for relaxed defects and their formation energies for all TM dopants. Given the limited research on WS_2, I also reviewed and drew insights from well-established studies on other material platforms for quantum defects, such as diamond and hBN, as well as the associated first-principles methodologies. These efforts culminated in my master thesis, including a comprehensive literature review and a detailed technical report. 这句写得好，明确地总结了这一段。

开始不太理解，院长的话-put it into practice-master programme做的东西，这三者之间的关系。（我本来以为院长的话是鼓励自学）

这个事例加的不错，一是表达你对NUS科研设施的认可，二是表达你有运用它们的技能。但这句话和前面一句话的逻辑关系突兀。

In practice

Training in machine learning While I possess the passion for physical and material science, I also maintain substantive interests in data science and artificial intelligence. I have pursued a minor degree in artificial intelligence and actively engaged in data-driven and AI projects. It is exciting to find that AI is being extensively used to empower and accelerate material discovery.

这句作为结尾，感觉下面你要开始讲AI和材料的结合了。但其实你没有讲，所以逻辑上有点不顺。

已调整

In the AI minor program at Shandong University, I learned about the algorithms and techniques that underlie AI capabilities, including machine learning, natural language processing, computer vision, and robotics. The substantial coursework made it demanding to balance with my major degree studies, as I

1）Training in machine learnign这个section展示了你丰富多面的尝试和能力，很不错。

这还不强

2）对于在山大辅修给你打的专业基础，虽然表达了，但似乎不够强。调整本section第一段这三句话和第二段第一句话，应可改善这个问题。

或许你是有意选择practice这个词（还有本段倒数第二行的my work），但根据你前两个sections的逻辑，会让读者误以为你这里是讲工作，但其实还是在讲学习经历。

只是在research和traning之外想不到第三个词了

often had to handle multiple projects on diverse topics simultaneously. Dealing with these challenges enabled me to cultivate useful skills in project and time management. My expertise in machine learning progressed further during my master's studies. I took *PC5252 Bayesian Statistics and Machine Learning*, where I acquired theoretical foundations from the Probabilistic Perspective. During *PC5251 Applied Machine Learning*, I delved into machine learning case studies within the physical science. Beyond my coursework, I spent three months learning and applying ensemble learning model to real-world datasets. During this exploration ~~time,~~ I successfully reproduced highly-rated public solutions and taught myself advanced ensemble techniques, as well as developing skills in feature engineering and hyperparameter tuning.

一些补充

这个方向，是与上面两个方向相关和延伸的呢，还是与要申请的phd方向相关的呢，还是第三个方向呢？上一次也讨论过，不记得当时的结论。

我难以理解这一段里事件之间的逻辑关系，一方面是语句之间因果或逻辑关系不明，也可能是因为我对专业专有名词的陌生无法简单get到。

Practice in numerical modelling My passion for applying computational methods to scientific challenges led me to explore various numerical modeling method. Motivated by multiscale modeling across different length scales, I chose finite element analysis (FEM) in Permanent Magnet Motors for my undergraduate capstone project in minor degree, where I gained basic knowledge in FEM analysis within Ansys Workbench. At NUS, inspired by *Prof. Wang Jian-sheng*'s detailed lectures on Monte Carlo (MC) methods, I used the MC methods and convolutional neural networks to identify phase transitions in 2D Ising model. In my work with Bayesian neural networks, I employed stochastic gradient Langevin dynamics in MCMC sampling as an optimization method for model parameters. 已调整

Research Interest My specific research interest is in using multiscale approaches combined with machine learning to discover and understand materials. I am particularly interested in three areas ~~of this research~~: 已改 Firstly, I plan to leverage machine learning to build efficient computational screening workflows, transforming the exhaustive search of all combinations in traditional routines. This includes utilizing transfer learning to take advantage of published databases and pre-trained models, and employing active learning to strategically explore potential chemical spaces, creating a modeling-search-computation iteration. Secondly, I am interested in applying interpretable machine learning techniques 已改

This指代不清，是指前面这一整句话？还是这句话里的某一个概念？对比一下第一个方向，this的使用似乎是不一样的。

in material science to extract physically meaningful insights. This involves both intrinsically interpretable models, such as SISSO and LASSO, and extrinsic explanation methods, like SHAP. Thirdly, I aim to utilize multiscale modeling approaches to capture the multiphysical mechanisms that govern material behaviors. This includes integrating microscale (DFT and molecular dynamics), mesoscale (phase-field method and force-field modeling), and macroscale (Finite Element Methods) approaches.

如果你这一段只是想表达较广发的兴趣，plan这个词用得不大恰当。用plan就好像在讲具体的研究计划和安排。第三的aim勉强可以，但也不是很恰当的。

本段重写

都包含

用词比较奇怪

未能理解这是指前面三个方向的结合，还是只是其中两个方向。

I believe NUS would cater well to my aspirations because of the diverse strength of its faculty and students. In the field of computational material science, I would be excited to work with XXX, XXX and XXX. Their professional expertise and research pursuits are what I have been eagerly anticipating. After completing my PhD, I would like to be active in material modelling community, whether it is in academia or industry. With the rapid development of HPC technology and AI capabilities, computer simulations will affect how materials science is performed. I believe that NUS physics department would be the ideal environment for me to continue my studies and become a valuable and contributing member.

这句话有歧义。如果是NUS物理系适合你读这个博，这句话应该写在本段前；你写在future career后，以为你是毕业想进NUS工作

如果你是想表达一个比较宽泛的向往，那就用such as；如果你想表达对某几个老师明确的兴趣，就需要表达他们具体什么让你感兴趣。只说professional expertise和research pursuits，很宽泛。

1）所以这里真的是把三个方向都结合起来了，不错！是真的是你想做的吗？
2）这一段的结构很不错，有三个你想做的事情，也紧跟有具体的解释，展示了你对于这个总体方向的构想，以及对基本知识的具备和灵活运用。
3）但是，你需要思考并充分表达这三个细分方向之间的联系。（我读后面的内容时，未能感觉像本段第一句话所说的那么清晰地相关。如果读博真的可以做，它们将会是三个独立的小项目，还是三个小项目之间是有什么紧密联系呢？如果是有紧密联系，文字上就需要表达，"第一想做什么，第二个想做的事情是解决第一个事情里的哪个环节的什么问题。"这种逻辑）
4）这三个领域/方向结合起来，有怎样的promising意义呢？可以在最后加一下。

这容易让读者觉得你没有清晰的方向，但是这也不是一个致命的问题。不过，还需要补充你对这两个方向的设想，各一两句话。跟后面那一句HPC和AI快速发展结合。